

MULTI-STAGE AI PROMPT WORKFLOW

AI Personalized Learning Platform

Assessment Submission

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1. Scenario Selected

The selected scenario is an AI Personalized Learning Platform. The objective is to transform a student's assessment and learning-performance information into a personalized learning plan, recommend suitable courses, generate topic-based quizzes, identify weak areas, and track progress.

The workflow is designed as a multi-stage AI process. The first stage organizes the requirements, the second stage converts that plan into a practical system design, and the third stage evaluates and refines the design.

2. Objective

- Assess a student's knowledge level across different topics.
- Identify weak topics from assessment and quiz performance.
- Recommend suitable beginner-friendly or level-appropriate courses.
- Generate AI-based quizzes for selected topics.
- Track individual course, topic, and quiz progress.
- Ensure each logged-in student sees only their own personalized information.
- Use a three-step chained AI workflow with a mandatory self-critique stage.

3. Input Data Used

The following sample student information is used to demonstrate the workflow. It represents the type of information that the platform can use for personalization.

Student ID	Topic	Assessment Score	Quiz Score	Current Level
S001	Python Basics	82%	78%	Intermediate
S001	SQL	58%	52%	Beginner
S001	Data Structures	64%	60%	Beginner
S001	Web Development	88%	84%	Intermediate
S001	Machine Learning	45%	40%	Beginner

4. Prompt Chain Transcript and Intermediate AI Responses

Step 1 – Extraction and Planning

Prompt 1 – Exact prompt used:

Act as an expert educational technology analyst and learning-platform designer. I will provide a student's topic-wise assessment and quiz performance.

Analyze the information and create a structured personalization framework. Identify:

1. Strong topics
2. Weak topics
3. The student's current learning level
4. Suitable course recommendations
5. Suitable quiz difficulty
6. A simple progress-tracking approach

Then create a structured plan for an AI Personalized Learning Platform with Student Login, Dashboard, Assessment, Courses, AI Quiz, Weak Topic Identification, and Progress Tracking.

Present the result using clear headings and a table. Keep the solution practical for a college project. The complete output will be used as the input for Step 2.

Intermediate AI Response – Step 1:

Personalization Analysis:

- Strong topics: Python Basics and Web Development.
- Weak topics: Machine Learning, SQL, and Data Structures.
- Current level: The student has mixed ability and should receive topic-specific recommendations instead of one overall difficulty level.
- Course recommendations: beginner-level Machine Learning and SQL, followed by Data Structures practice.
- Quiz difficulty: easier questions for weak topics, with moderate questions for stronger topics.
- Progress tracking: record course completion, assessment scores, quiz scores, weak-topic status, and improvement over time.

Platform Plan:

Student Login → Personalized Dashboard → Assessment → Topic-wise Analysis → Course Recommendation → Learning → AI Quiz → Result Analysis → Progress Update.

Step 2 – Generation / Drafting

Prompt 2 – Exact prompt used:

Using only the personalization framework and platform plan produced in Step 1, create a detailed system design for the AI Personalized Learning Platform.

Do not remove the Step 1 features. Transform the previous output into:

1. Website architecture
2. Login and student-account flow
3. Personalized dashboard
4. Assessment module
5. Course recommendation logic
6. AI quiz-generation process
7. Weak-topic identification
8. Progress tracking
9. Database tables
10. Frontend pages
11. Backend/API responsibilities
12. Development phases

Make the design realistic for a college project. Most importantly, explain how the platform ensures that the currently logged-in student sees that student's own courses, scores, quizzes, weak topics, and progress.

Intermediate AI Response – Step 2:

System Design:

- Architecture: Frontend → Backend/API → Database, with an AI service for quiz generation and recommendation support.
- Login: Each student signs in to a unique account. The authenticated student ID is used for student-specific data.
- Dashboard: Displays recommended courses, weak topics, recent quiz scores, assessment results, and progress.
- Assessment: Questions are grouped by topic and scored separately.
- Recommendation: Topic scores are compared with course difficulty to select suitable learning content.
- AI Quiz: The student selects a topic; the AI generates questions using the student's current performance and appropriate difficulty.
- Weak topics: Low assessment/quiz scores and repeated mistakes identify areas requiring more practice.
- Progress: Completed lessons, course status, quiz attempts, and score changes are stored and displayed.
- Database: Users, StudentProfiles, Topics, Courses, Assessments, Questions, AssessmentResults, Quizzes, QuizResults, and Progress.
- Frontend: Login/Register, Dashboard, Assessment, Courses, Course Details, Quiz, Results, Progress, and Profile.
- Backend: Authentication, student-data filtering, scoring, recommendation, quiz generation, result storage, and progress APIs.
- Development: Authentication/database → assessment → recommendation → quiz → dashboard → testing.

Step 3 – Evaluation and Refinement (Critique Gate)

Prompt 3 – Exact prompt used:

Before finalizing the system design, act as a strict senior reviewer. Evaluate the Step 2 design against this checklist:

1. Does it clearly personalize learning for each student?
2. Does the logged-in student see only that student's dashboard and learning data?
3. Does the assessment result directly influence course recommendations?
4. Is weak-topic identification clearly defined?
5. Is AI quiz generation included and connected to student performance?
6. Are the frontend, backend, and database components practical for a college project?
7. Are all important Step 1 features still present?
8. Is any part unnecessarily complex or unclear?

First list the weaknesses you find. Then produce a corrected final version that fixes the weaknesses without removing the core requirements from Step 1.

Intermediate AI Response – Step 3 Critique:

Review Findings:

- The design includes personalization, but the final version should explicitly require authenticated student filtering on every student-specific request.
- Assessment results influence recommendations, but the relationship should be stated as a direct rule.
- Weak-topic identification should use both low scores and repeated quiz mistakes.

- AI quiz difficulty should adapt to the student's current performance.
- The project is practical, but the first implementation should focus on the core modules rather than advanced infrastructure.

Refinement:

1. Add authenticated student ownership to all student-specific records.
2. Connect topic-wise assessment scores directly to course recommendations.
3. Define weak topics using low scores and repeated incorrect quiz answers.
4. Adapt quiz difficulty using the student's topic performance.
5. Keep the implementation modular and suitable for a college-level project.

5. Refined Final Output

5.1 Final System Workflow

Student Login → Student-specific Dashboard → Knowledge Assessment → Topic-wise Scores → Weak Topic Identification → Personalized Course Recommendation → Learning → AI-generated Quiz → Quiz Result → Updated Weak Topics and Progress → Personalized Dashboard

5.2 Main Modules

Student Login

Students register/sign in and the authenticated account controls access to that student's data.

Personalized Dashboard

Shows the logged-in student's assessment results, recommended courses, weak topics, quiz results and progress.

Knowledge Assessment

Measures topic-wise knowledge and stores results for the student.

Course Recommendation

Uses topic performance to recommend appropriate learning content and difficulty.

AI Quiz

Generates topic-based quizzes with difficulty adapted to the student's performance.

Weak Topic Identification

Marks topics with low performance or repeated mistakes for additional practice.

Progress Tracking

Stores completed lessons, quiz attempts, scores and improvement over time.

5.3 Database Design

Suggested tables: Users, StudentProfiles, Topics, Courses, Assessments, Questions, AssessmentResults, Quizzes, QuizResults, and Progress. Student-specific tables should contain the authenticated student's identifier so that the application retrieves and updates the correct account.

5.4 Expected System Output

- A personalized dashboard for each student.
- Topic-wise assessment performance.
- Identification of strong and weak topics.
- Recommended courses based on the student's current level.

- AI-generated quizzes with suitable difficulty.
- Quiz scores and improvement tracking.
- Progress information for the currently logged-in student.

6. Final Workflow Summary

Stage	Input	AI / System Action	Output
1. Extraction / Planning	Student performance + project idea	Identify strong/weak topics and create personalization plan	Personalization framework + platform plan
2. Generation / Drafting	Step 1 output	Create detailed website/system design	Operational system design
3. Evaluation / Refinement	Step 2 output	Review against checklist and correct weaknesses	Final polished platform design

7. Evidence of Live Conversation

The assessment requires direct evidence of the live conversation thread. The following ChatGPT shared conversation link provides access to the live three-step AI workflow used for this assignment.

ChatGPT Conversation Link: <https://chatgpt.com/share/6a9ec4be-9474-83ee-8dcb-410ec9834d5d>

The shared conversation contains the chained prompts and responses used for the workflow.

8. Conclusion

This multi-stage AI workflow demonstrates how an AI Personalized Learning Platform can be designed through chained prompting rather than a single request. The first stage structures the student's learning requirements, the second stage converts them into a practical system design, and the third stage critically evaluates and improves the design. The final system connects assessment, personalized recommendations, AI-generated quizzes, weak-topic identification, and individual progress tracking.